

SaiNithinGoud Kurremula

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SUMMARY

Software Development Engineer skilled in full-stack development, cloud-native systems, and scalable microservices. Proficient in Python, Java, React, and AWS, with a track record of building responsive applications and high-performance REST APIs. Strong foundation in algorithms, data structures, and object-oriented design. Passionate in delivering innovative, user-centric software solutions, with emphasis on scalability, clean architecture, and system performance.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, C++, C, SQL, TypeScript
- **Frameworks & Libraries:** React, Django, Flask, FastAPI, Spring Boot, Node.js, Tailwind CSS
- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, XGBoost, RAG Pipelines, Multi-Agent Systems
- **Databases & Data Systems:** PostgreSQL, MySQL, MongoDB, Redis, Vector Databases
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD Pipelines, Kafka
- **Tools & Practices:** Algorithms, Data Structures, Git/GitHub/GitLab, Microservices, REST & GraphQL APIs, Postman, Observability

EXPERIENCE

SkyinfoLab Software Solutions

Jan 2024 - Jun 2024

Software Engineer

India

- Engineered scalable web applications with Python/Django, implementing modular architecture that improved maintainability and reduced response times by ~30%.
- Architected and optimized RESTful APIs, applying caching and query optimization that reduced API latency by 25% and improved throughput under peak loads.
- Built and deployed scalable backend APIs serving thousands of requests/day with <200ms latency.
- Developed responsive React UIs with async data fetching and state management, reducing page load times by 20% and boosting user engagement.
- Implemented optimized PostgreSQL and MongoDB architectures with indexing and partitioning, cutting query execution time by 35%.
- Collaborated with cross-functional teams in an Agile environment to debug production issues and utilized Git and CI/CD pipelines for seamless version control and automated deployments, improving deployment efficiency and reducing downtime.

PROJECTS

E-commerce & Real-Time Analytics Platform

- Built a microservices-based marketplace (Node.js) behind an API gateway with JWT + RBAC; supported REST + GraphQL, responsive React frontends, and async state management, reducing page load by 20%.
- Engineered Kafka topics for orders, payments, and inventory using partition-affinity + back-pressure (bounded queues), boosting throughput +40% under peak load.
- Designed a graph-based recommendation engine (adjacency lists + BFS/Dijkstra) with Redis/LRU caching, cutting stock mismatches and lifting conversions.
- Optimized query paths with materialized views + inverted indexes in PostgreSQL, reducing checkout latency by 35%.
- Established CI/CD pipelines (Terraform) for ECR builds + EKS rolling deploys; improved reliability with idempotency keys + DLQs on Stripe webhooks to guarantee exactly-once order transitions.

MoneyScale – Smart Finance Management App

- Built a budgeting assistant with a multi-agent pipeline (forecasting + anomaly detection + recommendations) using LangChain, improving categorization accuracy by ~20%.
- Combined time-series models (Prophet/ARIMA) with LLM simulations to generate personalized savings insights, increasing forecast accuracy to ~90%.
- Designed a secure, scalable backend with OAuth2.0, PostgreSQL (indexed/materialized views), and a serverless ETL (Lambda + S3) for receipts and reports.

AI-Powered Resume Evaluator

- Deployed auto-scaling Flask APIs with queue-based load balancing, sustaining p95 latency under 250ms across live evaluations.
- Orchestrated an LLM scoring pipeline with fairness checks, drift detection, and MLflow experiment tracking, ensuring consistent and unbiased candidate evaluation.
- Persisted results in MySQL with idempotent job IDs, tracked experiments in MLflow, and built a React dashboard for insights.

DDoS Detection

- Designed pipeline: parsing & cleaning flow features (80+), scaling, feature selection, and SMOTE for imbalance.
- Trained RF/XGBoost ensemble to 98%+ F1; tracked ROC-AUC, precision/recall; threshold tuning per attack type.
- Packaged the model as a reproducible scikit-learn Pipeline with versioned artifacts; deployed in shadow mode to compare against baseline traffic before enforcing blocks.

EDUCATION

University of Cincinnati

Aug 2024 - Present

Master's, Computer Science

- **Coursework:** Algorithms & Data Structures, Distributed Systems, Web/Cloud Computing, Database Systems, Machine Learning